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A SECOND VIRTUAL POLE FROM CORSICA, THE OTA GABBRODIORITE[1]

A. E. M. NAIRN AND M. WESTPHAL

Department of Geology, Western Reserve University, Cleveland, Ohio (U.S.A.); Institut de Physique du Globe, Strasbourg (France)

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SUMMARY

Palaeomagnetic measurements on a series of gabbrodiorite samples from restricted outcrops around Ota in northwest Corsica are described. They yield a virtual pole position in 53.5 N, 173.3 E. An attempt is made to assess the significance of this result.

INTRODUCTION

The mountainous island of Corsica lies in the Gulf of Genoa in the angle formed by the coasts of southern France and Italy. The main geological feature of the island, a granite massif lying to the south and west of a line running roughly from Ile Rousse to Solenzara with schists to the north and east (Fig.1), has been recognised ever since the earliest geological researches of HOLLANDE (1877). If the granites have been generally regarded as Hercynian in age (BROUWER, 1957; DENIZOT, 1958) there is less agreement about the age of the schists which have been variously regarded as Tertiary by some authors and as Hercynian or older by others (see LAPADU-HARGUES and MAISONNEUVE, 1963). Between the two lies a narrow irregular zone in which Mesozoic sediments are found.

Lying unconformably upon the granites are Permo-Carboniferous sediments and volcanics. On the Osani peninsula the oldest sediments appear to be Late Westphalian or Early Stephanian in age, but near Argentella Dinantian is known (KRYLATOV, 1963). It was upon the basis of a preliminary palaeomagnetic survey of the Permo-Carboniferous rhyolites or welded tuffs that ASHWORTH and NAIRN (1965) suggested a tectonic rotation of Corsica. Detailed study of these rocks is in progress.

Within the main granitic zone various bodies of dioritic and gabbroic rocks

[1] Contribution No. 14.1, Department of Geology, Western Reserve University, Cleveland, Ohio.

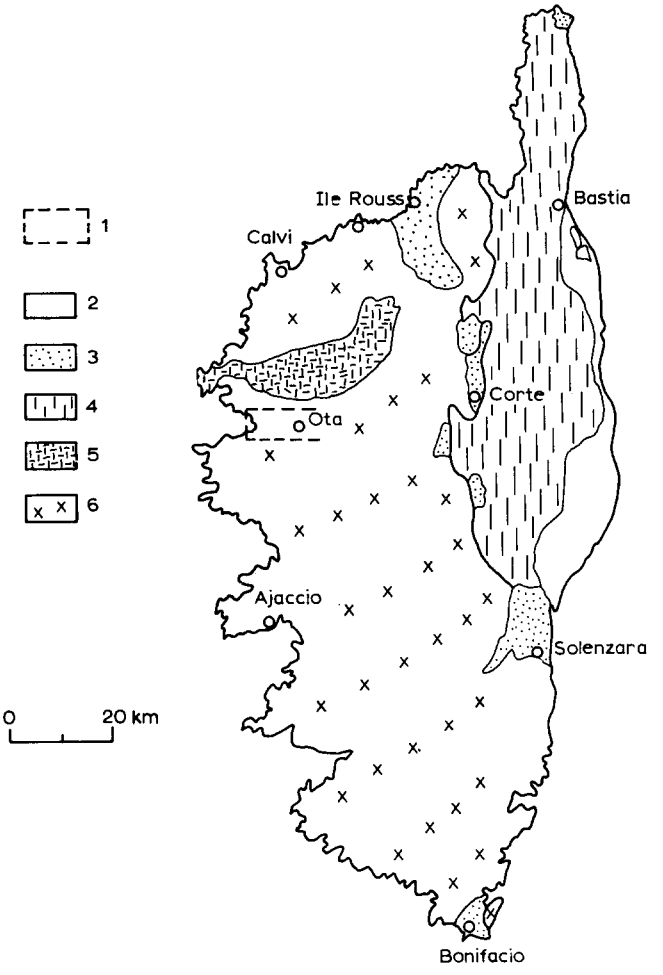


Fig.1. Outline map of Corsica showing the main geological divisions of the island and the location of the area sampled.

Legend: *1*=area sampled enlarged in Fig.2; *2*= Quaternary deposits; *3*=Mesozoic and Tertiary basins; *4*=schistes lustrées; *5*=Permian volcanics; *6*=granite of the western massif.

occur such as the massif forming the Iles Sanguinaires, and the famous orbicular diorite of Sartène and the gabbrodiorite of Porto-Ota-Evisa with which the present note is concerned. The age of these rocks is in general poorly known.

THE GABBRODIORITE OF OTA

In the Porto valley on either side of the village of Ota are good but restricted outcrops of gabbroic rocks exposed in fresh road cuttings, elsewhere outcrops are rare being either concealed in the macquis or under a cover of "eboulis". They

form a narrow zone within the biotite granite massif, having an area of approximately 6 km^2. There are two interpretations of the age relationship of these basic and acidic rocks. DEPRAT (1905, p.46) summarily regarded the basic rocks as post granitic intrusions of probably the same age as the diabase dykes found in the Osani area. This view, unsupported by any evidence is that represented on the geological map of the area (DEPRAT, 1906) and is shown in Fig.2A. Recently TELLINGEN (1955) has expressed the view that the basic rocks form a zone of enclaves within the granite (Fig.2B) of which the larger are prominently displayed in road cuttings. This interpretation is based upon detailed observations on the smaller enclaves a few metres in extent which can be studied in their entireity. They are in some places so numerous that they are locally more important than the granite. They are traversed not only by granitic veins, but also by pegmatite and aplite veins. The margins of some of the enclaves may show slight alteration, and fluidal structures can sometimes be found in the enclosing granite. Measurements of the joint plane directions in the larger gabbrodiorite outcrops gave stereograms very

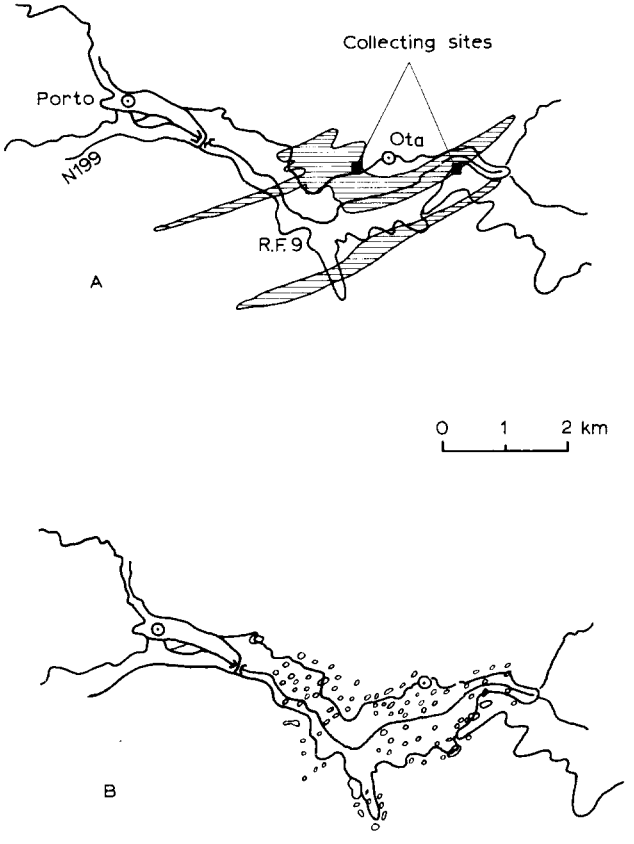


Fig.2A. The outcrops of the basic rocks near Ota according to the geological map.
B. The distribution of the outcrops of the same basic rocks according to TELLINGEN (1955).

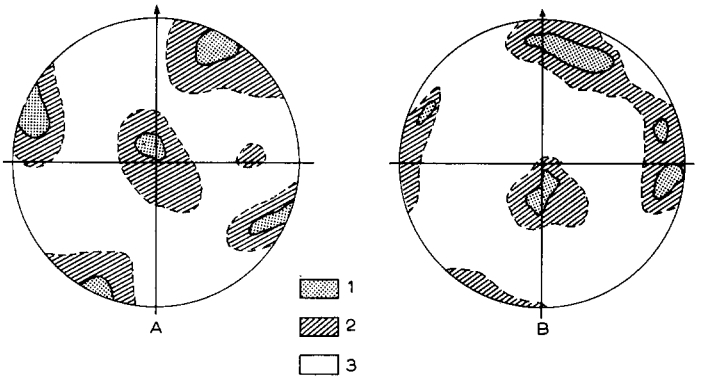


Fig.3A. Plot of the joint plane measurements in the gabbrodiorites (about 200 measurements). B. Plot of the joint plane measurements in the granite (about 300 measurements). Legend: *1* = >4 points per 15 °square; *2* = 4 points per 15 °square; *3* = 2 points per 15 °square.

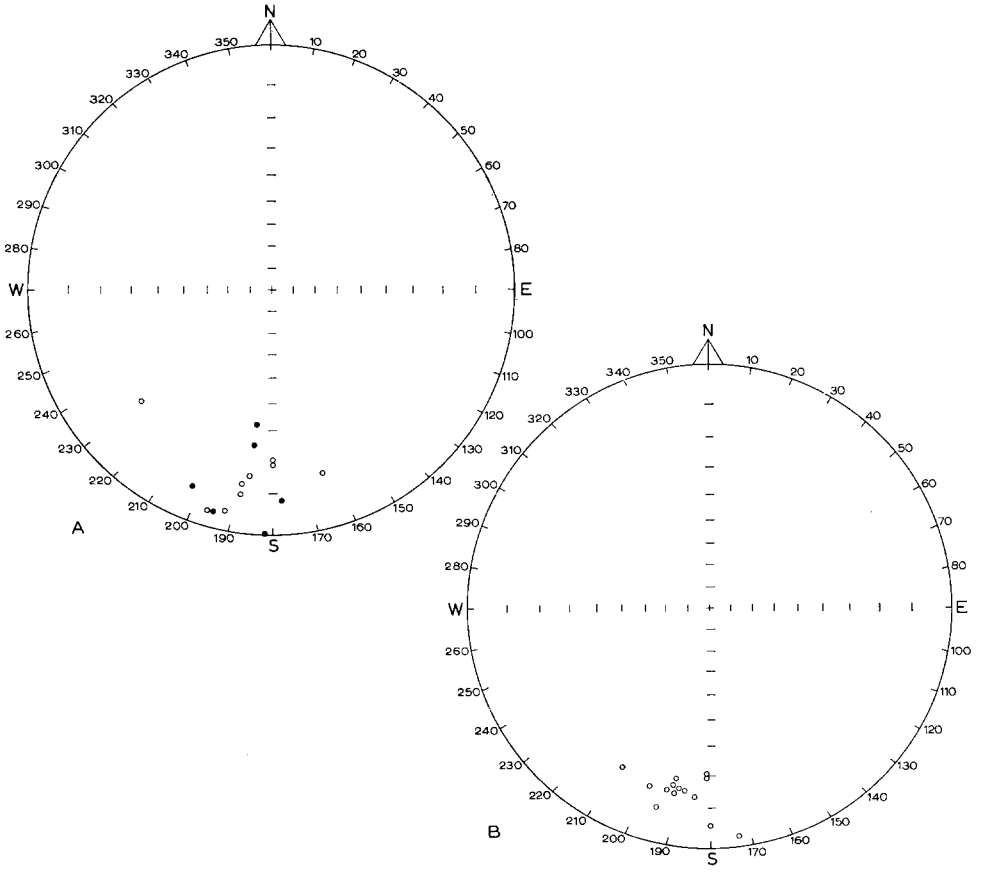


Fig.4. The directions of magnetization of the gabbrodiorite near Ota. A. Natural remanent magnetization. B. After demagnetization in a field of 170 Oe.
Legend: ● = positive inclinations; ○ = negative inclinations.

little different from those obtained from like measurements in the granite (Fig.3).

PALAEOMAGNETIC OBSERVATIONS

Sixteen samples were collected at two sites about 2–3 km apart on either side of the village of Ota. At both sites fresh surfaces were exposed in road cuttings and sampling was spread over a lateral distance of at least 100 m of outcrop in each case. These outcrops were not unbroken, and in TELLINGEN's (1955) sense would represent several enclaves. From each hand sample a single 2.5 cm core was prepared on which measurements of direction and intensity of magnetization and alternating field demagnetization tests were carried out. Only two specimens proved to be insufficiently magnetic to give results under the short period low sensitivity astatic magnetometer used. Progressive demagnetization was carried out in fields of up to about 700 Oe with little change in direction. The best representation of the recorded direction of magnetization was regarded as that in which the mean value was associated with the maximum value of k (FISHER, 1953). This was found after demagnetization in fields of 170 Oe. The directions of natural remanent magnetization and after demagnetization are shown in Fig.4. Typical demagnetization curves are given in Fig.5. The changes in the direction of magnetization during the demagnetization process were not great as a comparison of Fig.4A, B shows, the mean value of NRM was $186.9-1.9\ \alpha = 9.4$, and after 170 Oe, $189.2-13.1\ \alpha = 5.4$.

TABLE I

SUMMARY OF DATA

Ota, geographic coordinates	42.2N, 8.8E
Mean direction of magnetization	189.2–13.1
Circle of confidence	5.4°
k	55.6
Number of samples in computation	14
Number of samples excluded	2 (too weakly magnetized)
Demagnetizing field	170 Oe
Mean intensity of magnetization (NRM)[1]	$1.69 \cdot 10^{-4}$ e.m.u./cm^3
Mean-standard deviation/mean+standard deviation	$0.4/6.9 \cdot 10^{-4}$ e.m.u./cm^3
Mean intensity of magnetization (after 170 Oe)	$1.75 \cdot 10^{-4}$ e.m.u./cm^3
Mean-standard deviation/mean+standard deviation	$0.4/7.3 \cdot 10^{-4}$ e.m.u./cm^3
Mean susceptibility	$3.82 \cdot 10^{-4}$ e.m.u./cm^3
Mean-standard deviation/mean+standard deviation	$1.7/8.7 \cdot 10^{-4}$ e.m.u./cm^3
Virtual pole position	53.5N, 173.3E
Virtual pole position after rotation	24.9N, 113.2E

[1] Log. mean, mean—standard deviation, mean+standard deviation are computed and are given for convenience in antilog. form.

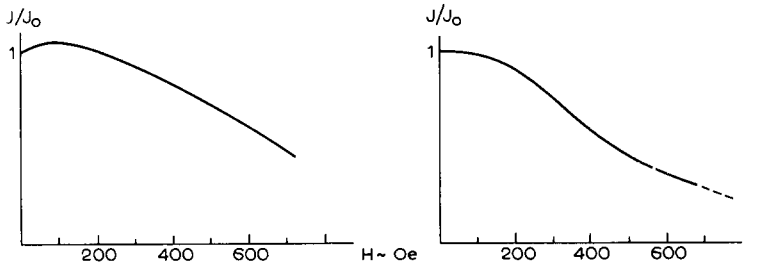


Fig.5. Normalized demagnetization curves of the Ota gabbrodiorite in alternating fields. Vertical axis normalized intensity, horizontal axis field intensity in oersteds.

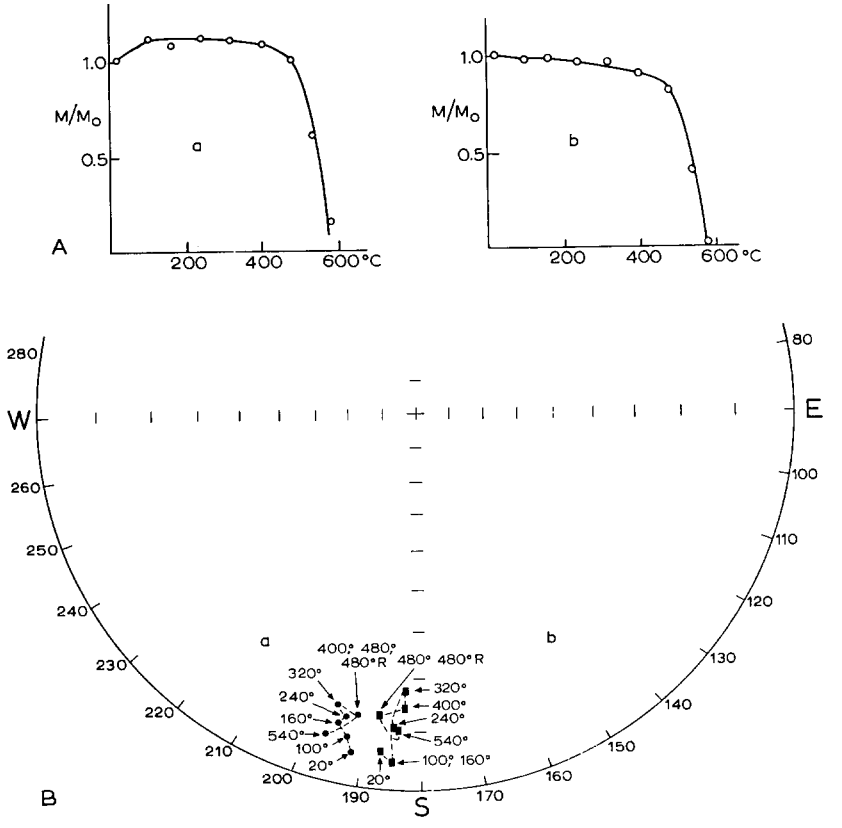


Fig.6A. Normalized thermal demagnetization curves of two Ota gabbrodiorite samples. Vertical axis normalized intensity, horizontal axis temperature in °C.

B. The changes in direction during demagnetization. All inclinations are negative.

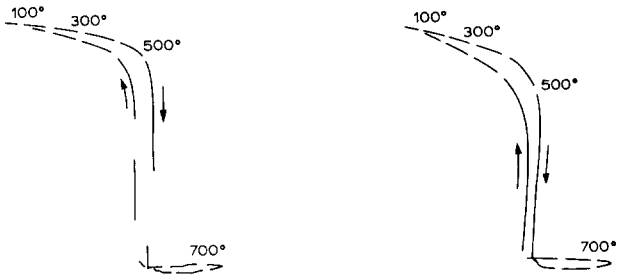


Fig.7. Thermomagnetic curves of the Ota gabbrodiorite. The curves are copies of the photographic records obtained from a Curie–Chevallier balance.

An additional core from every fourth sample was thermally demagnetized and two representative curves are shown in Fig.6A. Apart from indications of a comparatively small viscous component, the magnetization appeared stable, the changes in direction during demagnetization are given in Fig.6B. Only a single Curie point about 580° was observed. Thermomagnetic curves, of which two copies of the photographic record obtained are illustrated in Fig.7, were obtained using a Curie–Chevallier balance. They show a small shift in the Curie temperature in the cooling curve only a little greater than the thermal lag of the instrument and suggest that the ferromagnetic component must be close to pure magnetite. Susceptibility was measured in fields of the order of 2 Oe using a susceptibility bridge (COLLINSON et al., 1963).

In computing mean values all samples were combined as if from a single site, as it appeared from DEPRAT's (1905) map, upon which collecting was planned, that further outcrops would not be readily accessible. Because of the geological history of the gabbrodiorite this mean direction of magnetization cannot necessarily be considered as original (see following section). From the mean direction a virtual pole position was computed, and this, along with a summary of other statistics is given in Table I. The mean intensity and susceptibility are calculated in log. form, the antilogs of which appear in the table.

DISCUSSION OF RESULTS

There are three main problems to consider when attempting an interpretation of the measurements: (*1*) the age of the gabbrodiorite and granite, (*2*) the age of the magnetization, and (*3*) questions arising out of the restricted sampling.

(*1*) If TELLINGEN (1955) is correct, as seems reasonable on the basis of his description, then the gabbrodiorite is older than the granite. According to DEPRAT (1905) it is younger. As the gabbrodiorite is found wholly within the granite, the basis of any estimate of geological age depends upon what is known of the host rock. The granite some 20 km to the north of the Ota region is overlain by Upper

Carboniferous sediments (Upper Westphalian or Stephanian), and near Argentella Dinantian rocks are found (KRYLATOV, 1963). That the granite below the unconformity does not appear highly weathered or that there is no basal conglomerate as age information is inconclusive. It is only possible to state that the granite cannot be younger than Early Carboniferous, the upper age limit is unknown.

(*2*) On the basis of TELLINGEN's (1955) conclusions it is necessary to consider the thermal history of the rock to determine whether the heating which must inevitably have resulted exceeded the Curie temperature. Use can be made of THELLIER's (1941) observation that rocks retain a "memory" of thermal events subsequent to their formation. Thermal demagnetization, however, reveals evidence of only a single thermal event, and as it is inconceivable that the gabbrodiorite as enclaves could have escaped heating, it would seem that they were heated above their Curie temperature with the consequence that the thermoremanence recorded is that acquired during the cooling of the granite.

(*3*) The restricted sampling may affect the interpretation in two ways, by

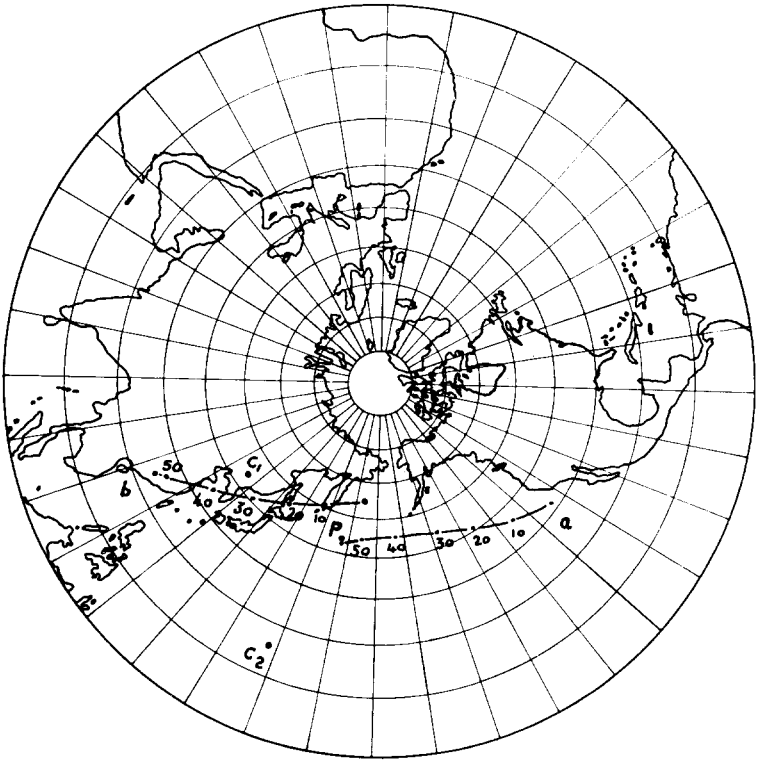


Fig.8. The virtual pole position of the Ota gabbrodiorite and its change (curve *b*) assuming the rotation proposed on the basis of the preliminary survey of the Osani rhyolites (curve *a*). Points C1 and C2 represent the mean directions of the two Carboniferous groups given by IRVING (1964), P is the mean Permian pole from northern and central Europe.

making it difficult to assume that the effects of secular variation have been averaged out, and by giving little control over the local tectonic environment. If the arguments for the age of magnetization on TELLINGEN's (1955) theory are feasible, then the comparatively slow cooling of the granite make the secular variation assumption not improbable. If DEPRAT (1905) is correct, then only the distribution of sampling can give any indication of a sufficiently long time coverage. There is no direct tectonic control within the area sampled or of its relations to other parts of the granite massif. It can only be remarked that in the overlying Upper Palaeozoic sediments there is evidence of only minor faulting, and that there are no obvious tectonic lines breaking through the massif, nor have any been proposed. The only gross tectonic effect so far proposed is the tentative assumption of a rotation of the massif as a whole (ASHWORTH and NAIRN, 1965).

In the absence of any indications to the contrary, it will be assumed that the conditions discussed in (*3*) above are met. There are then two possible interpretations of the virtual pole position (53.5N, 173.3E) computed, that based on Tellingen and that following Deprat.

From the first it would appear that the virtual pole position is anomalous for it differs from all known virtual pole positions from European rocks of Early Carboniferous age or older. If the rotation hypothesis, as based on the tentative interpretation of Permian results (Fig.8, curve *a*) is applied, then the virtual pole position is displaced along curve *b* (Fig.8) to a position 24.9N, 113.2E which is fairly close to one of BUCHA's (1964) Silurian poles (39N, 113E). The uncertainties both geological (of the exact granite age) and palaeomagnetic preclude any more definite conclusion. The dispersion of the Palaeozoic pole positions excluding those of Permian age is great, IRVING (1964) found it convenient to compute two virtual pole positions for Carboniferous rocks shown as points C_1, and C_2 in Fig.8, whilst CHAMALAUN and CREER (1963) have recently radically revised the generally accepted Devonian pole position.

If DEPRAT (1905) is correct then the gabbrodiorite must be younger than the granite, and on the basis of his equating its intrusion with the Osani diabase dykes, must be younger than the Permian volcanics. On this basis the virtual pole position is not anomalous, would suggest a Permian age and rule out the rotation hypothesis as proposed.

CONCLUSION

There are two possible interpretations of the measurements made on the Ota gabbrodiorites, and the uncertainties, both geological and geophysical, preclude the decisive rejection of one. It is the authors' opinion that the interpretation based upon TELLINGEN's (1955) work is the better, since his descriptions seem to clearly indicate the older age of the gabbrodiorite. This interpretation favours

rotation. However, DEPRAT's (1905) work would indicate a contrary result which would render rotation unnecessary.

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